Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under Section 2(f) of UGC Act, 1956

End Term (Even) Semester Regular/Back Examination May 2026

Roll no................................

Name of the Program and semester: B.Tech./VIth
Name of the Course : Software Engineering
Course Code: TCS611
Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1.

(2X10=20 Marks)

a. A project consistently exceeds deadlines and budgets despite skilled developers. What underlying issue does this indicate? CO1

b. Critically analyze the applicability of the Waterfall model and prototype model in modern software development environments. Discuss its strengths and limitations. CO3

c. A project emphasizes strict documentation and rigid planning but fails to adapt to changing requirements. What does this indicate about its alignment with Agile values? CO1

Q2.

(2X10=20 Marks)

a. A company is developing an online shopping platform. Initially, the focus was on basic product listing and ordering, but stakeholders later demand features such as personalized recommendations, multiple payment options, and real-time order tracking. Frequent requirement changes are causing delays. Analyze the challenges faced in requirement engineering in this scenario and propose a structured approach for requirement elicitation and analysis. CO5

b. A Hospital Management System enables patient registration, appointment scheduling, and billing. It must also maintain data privacy and provide quick access to patient records. Identify and differentiate functional and non-functional requirements with suitable examples. CO2

c. Hospital Diagnosis Support System determines treatment options based on symptoms, test results, and medical history. Explain how Decision Trees and Decision Tables can represent medical decision logic, and how Document Flowcharts help in understanding data flow. CO3

Q3.

(2X10=20 Marks)

a. A Library Management System has separate modules for book issuing, returning, and catalog management, but these modules frequently depend on each other's internal data structures. Evaluate the system design in terms of coupling and cohesion and suggest how modularity can be improved. CO4

b. An Online Payment Processing Module handles multiple conditions such as valid card, insufficient balance, expired card, and OTP verification. A tester constructs a Control Flow Graph

(CFG). Explain how cyclomatic complexity is calculated and how CFG helps in identifying independent paths for testing. CO6

c. A Ride-Sharing Application Development Team faces issues with code readability, inconsistent formatting, and poor documentation, affecting scalability and maintenance. Discuss how adopting coding standards, writing clean code, and maintaining proper documentation can resolve these issues. CO5

Q4. (2X10=20 Marks)

a. A Payroll Management System calculates salaries, deductions, and generates reports. Each module is tested separately, integrated, and tested as a whole. The system is verified by management and later modified, requiring regression testing. It is also tested for accuracy and performance. Analyze the different types of testing performed. CO6

b. A banking transaction system is evaluated using peer reviews, walkthroughs, code inspections, and automated tools before execution. Examine, analyze, and evaluate different static testing strategies including Formal Technical Reviews, Walkthroughs, Code Inspection, compliance with standards, and automated testing tools. Propose how these techniques improve software quality. CO1

c. A Social Media Application is internally tested by developers for functionality and performance, and later released as a beta version to a group of users for feedback and bug reporting. Explain Alpha Testing and Beta Testing with examples from this system. CO3

Q5. (2X10=20 Marks)

a. A financial system project of size 80 KLOC is categorized under Organic mode. Apply and analyze the Basic COCOMO model to calculate and interpret:

1. Effort (person-months)
2. Development time (months)

Average team size Also, evaluate how estimation impacts project planning. CO4

b. A product-based company wants to enhance its software quality and achieve global standards certification. It decides to implement a maturity model. Analyze the role of SEI-CMM in achieving quality benchmarks. Propose a roadmap and evaluate how each maturity level contributes to continuous improvement. CO6

c. Write short notes on any two of the following: CO2

i. Types of maintenance
ii. CASE Tools
iii. Software Risk Management
iv. Software Configuration Management